Accuracy as a Fraction

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

Teach each score as an answer to a stated counting question.

Introduce matrix terms here, provide complete scripts and prediction records, and preserve the poll extension as a separate teaching cycle.

Keep one habit: Require raw counts and both class denominators before accepting a headline score.

MAIN LESSON · SECTION 1

Prepare the accuracy lesson and its denominator extension

> Separate the main assessment from the optional poll investigation.

Primary target and preparation

Primary target: Explain why overall **accuracy** can hide failure on a less common class

Begin with 9/12 accuracy and teach the matrix layout in this week. Do not require the future error-type lesson as a prerequisite. Prepare twenty fictional case counters, the supplied prediction table and blank labelled matrices. The poll is a connected denominator extension:

retain its five tasks and schedule separate time rather than treating its responses as a second copy of the classification assessment.

Teach two 45-minute sessions. Begin with a majority guess that looks successful under overall **accuracy**, then keep both class **denominator**s visible.

Introduce **balanced accuracy** as an equal average of two correct fractions, without treating it as a universal replacement for task-specific evaluation.

By the end, I can compare **accuracy** with class-specific performance in an imbalanced evaluation.

Materials and retrieval

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer **count**ers or a calculator after learners show the setup. Fictional records are supplied; personal data and paid ac**count**s are unnecessary.

Without opening the old lesson, explain one key idea from **Training** and Testing. Give an example and a mistake that the idea helps you avoid.

Look for: Choose a split unit that matches the intended **prediction** claim

Reference example, not the only acceptable wording: **Training** fits the model; validation compares development choices; test evaluates after choices are fixed.

Repeatedly selecting by the test score uses that set for development, so it is no longer an untouched final evaluation.

Without opening the old lesson, explain one key idea from Features and Distractors. Give an example and a mistake that the idea helps you avoid.

Look for: Choose recorded properties that preserve distinctions needed for a stated task

Reference example, not the only acceptable wording: Use length and width measured before the decision. State the order as length in cm, then width in cm; the record is 12, 8.

The target is fits or too large under the guide, and this card is fits because both equalities are included.

Colour and card number are not needed for this rule; the final inspection sticker copies the answer and is unavailable before inspection.

Without opening the old lesson, explain one key idea from Debugging Strategies. Give an example and a mistake that the idea helps you avoid.

Look for: Design revealing tests for a function

Reference example, not the only acceptable wording: 4, 0 and 6.

Printed lesson and workbook

Twenty **count**ers with sixteen labelled clear and four blocked

Blank confusion matrices and fraction strips

Optional Python 3 editor with no packages required

Flexible session plan

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

▼ Optional poll preparation and readiness

Schedule the connected poll extension separately when useful. Use fictional Yes/No/Unknown responses, label known and all-response **denominator**s, and retain its five workbook tasks and separate assessment. It is part of this week's retained content, not a prerequisite from Week16.

3/12, which is 1/4.

Python 3 editor with a fresh local practice file

Paper trace table and the student workbook

Fictional **input** records from this lesson; no real student identifiers

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Case exposure check: this source example appears in ai-2/12, ai-4/16. Reuse a familiar case for recall or modelling; verify that the case used to assess transfer has not already received feedback.

Retrieval answer guidance

Training and Testing

Features and Distractors

Debugging Strategies

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Class A gets 8/10 cases right and class B gets 18/30 right. Compute each rate and the combined rate. Why is averaging the two **percentage**s misleading here?

Concept to explain

Accuracy is correct **prediction**s divided by all **prediction**s.

Term	Meaning
accuracy	The number of correct predictions divided by the total number of scored predictions.
numerator	The top number of a fraction, identifying the counted portion.
denominator	The bottom number of a fraction, identifying the number of equal parts or total cases considered.
percentage	A proportion expressed per hundred; multiply a fraction by 100 to express it as a percent.

Prepare the class

This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Readiness answer

They are in the blocked-reference row and clear-**prediction** column: the lower-left cell. These are missed blocked cases.

Apply the connected idea: Summaries with counts and proportions

Readiness answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Model a fraction, then distinguish the two class denominators

> Supply complete code and introduce the matrix notation here.

Accuracy and matrix orientation

Accuracy counts all correct **prediction**s and divides by all evaluated examples. Each example has equal weight, so a common class contributes more to the score. A high result can hide failure on a rare class.

This week keeps the confusion matrix visible and checks both classes separately.

Read the matrix like a labelled counting grid. Each row names the reference class; each column names the predicted class. Here both axes are clear, then blocked. Clear→clear and blocked→blocked are correct. Clear→blocked is a false positive (a false alarm); blocked→clear is a false negative (a missed blocked case). Positive means the class we choose to track, not good or desirable. We introduce these names here; no future lesson is a prerequisite.

Read the main baseline and candidate counts

New fictional **training** data has 16 clear and 4 blocked labels. Its **majority baseline** predicts clear for every query; choose that rule from **training**, not evaluation labels.

A separate evaluation set also contains 16 clear and 4 blocked cases. A frozen candidate and the baseline are evaluated on the same 20 records.

Rows are references and columns are **prediction**s, ordered clear then blocked. Blocked is positive. Both methods get 16/20=80 percent **accuracy**. The baseline misses all four blocked cases.

The candidate finds all four but falsely flags four clear cases. The same score conceals a different tradeoff.

Predict and execute the complete scoring script

```
def scores(counts):
    clear_n = sum(counts[0])
    blocked_n = sum(counts[1])
    if clear_n == 0 or blocked_n == 0:
        raise ValueError("Need evaluation examples of both classes")
    correct = counts[0][0] + counts[1][1]
    accuracy = correct / (clear_n + blocked_n)
    clear_fraction = counts[0][0] / clear_n
    blocked_fraction = counts[1][1] / blocked_n
    balanced = (clear_fraction + blocked_fraction) / 2
    return accuracy, balanced

baseline = [[16, 0], [4, 0]]
candidate = [[12, 4], [0, 4]]
for name, counts in [("baseline", baseline), ("candidate", candidate)]:
    accuracy, balanced = scores(counts)
    print(name, round(100 * accuracy, 1), round(100 * balanced, 1))
```

Explain the two returned scores and their limits

The **output** is baseline 80.0 50.0, then candidate 80.0 87.5. In each **output** line the first **percentage** is ordinary **accuracy** and the second is **balanced accuracy**.

For the candidate, the clear-class correct fraction is 12/16=75 percent and the blocked-class fraction is 4/4=100 percent. Their equal average is 87.5 percent.

For the baseline, those fractions are 16/16=100 percent and 0/4=0 percent, averaging 50 percent. The code assumes a two-by-two table of nonnegative integer **count**s in the stated order.

It stops when either reference class is absent. Then report the observed **count**s and the unavailable class result; do not invent a **denominator** or silently substitute zero.

Collect the equal-score explanation

Same overall score different class results

SUMMARY	ALWAYS CLEAR BASELINE	FROZEN CANDIDATE
Overall correct	16 of 20 is 80 percent	16 of 20 is 80 percent
Reference clear correct	16 of 16 is 100 percent	12 of 16 is 75 percent
Reference blocked correct	0 of 4 is 0 percent	4 of 4 is 100 percent

Overall accuracy weights classes by their number of evaluation examples. Balanced accuracy gives the two class-specific fractions equal weight.

Both methods score 80 percent. Explain which cases the baseline fails, how the candidate changes the errors, and why their balanced accuracies differ.

Balanced accuracy is a metric choice, not a repair to the classifier or a guarantee of safety. It gives classes equal influence in this summary; it does not encode the actual consequences of false alarms and missed cases.

Four blocked evaluation examples are also too few for a strong general claim.

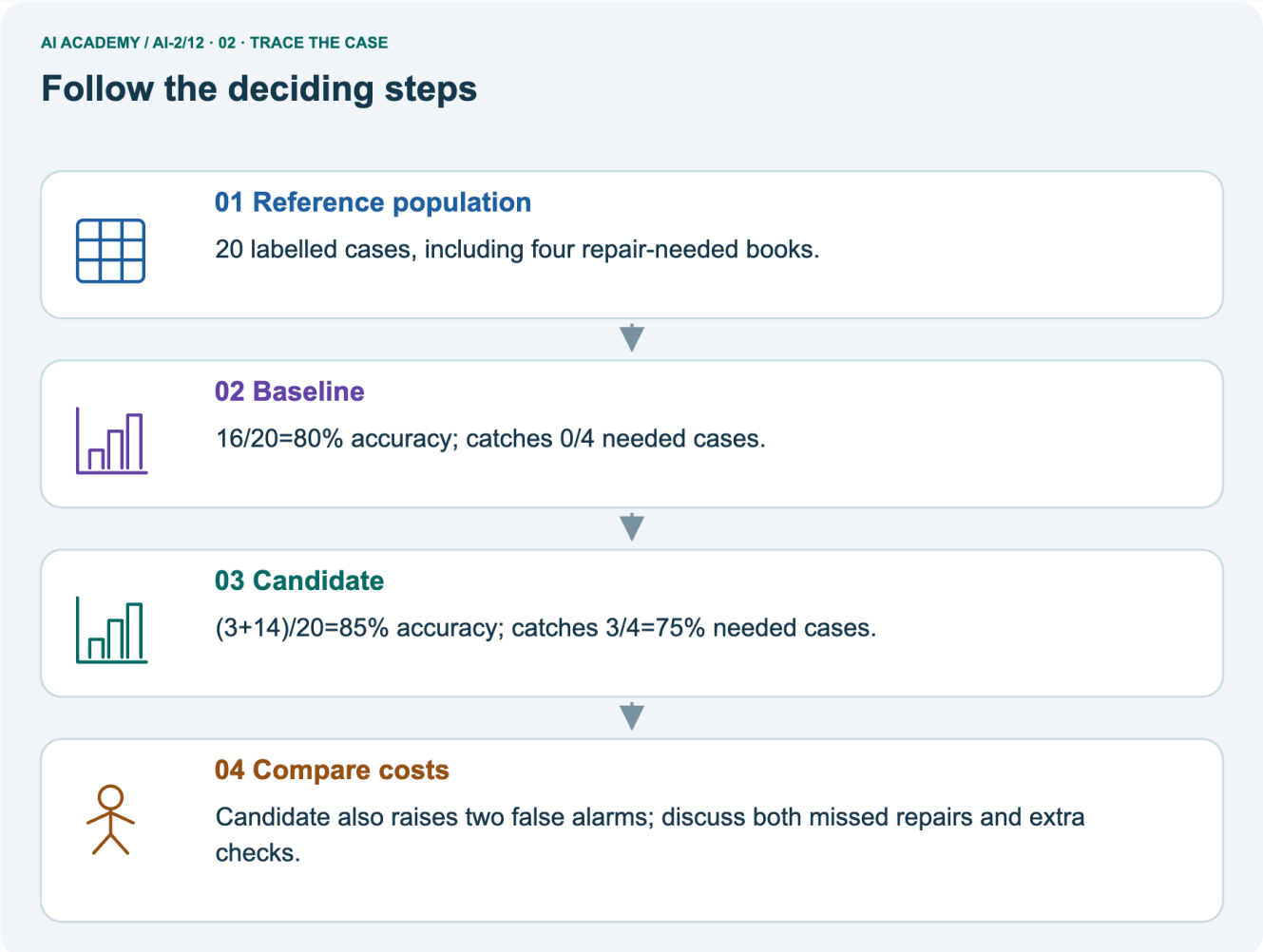
Introduce the separate shelf counts

A fictional shelf classifier predicts repair needed or not needed.

Evaluation reference labels: 16 not-needed and 4 needed. Positive class is needed.

Always-not-needed baseline is correct on 16. Candidate has TP=3,FN=1,FP=2,TN=14.

Trace detection and false alarms



Follow the labelled picture with the matching explanation. Open the vector diagram

A majority-class baseline can look accurate while missing every positive case. The candidate improves overall agreement and repair detection in this table, but false alarms still matter.

Use the same cases, **denominator**s and positive-class definition for both. Do not infer that a single metric identifies the best policy for every library.

These selected cases do not establish future rates or specify the real cost of each error.

Interpret the always-repair-needed change

Change: A candidate instead marks every book repair needed.

Prediction: Recall is 4/4=100%, but **accuracy** is only 4/20=20%.

Reason: All 16 not-needed books become false alarms.

Use the picture actively

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

▼ Optional poll model and key

A fictional poll asks whether a short activity was useful. Responses are Yes, No or Unknown. We will report Yes among known responses and also disclose unknown responses.

Choosing a **denominator** changes the question being answered.

```
responses = ["Yes", "No", "Yes", "Unknown", "Yes"]
yes = 0
known = 0
for response in responses:
    if response == "Yes" or response == "No":
        known = known + 1
        if response == "Yes":
            yes = yes + 1
if known > 0:
```

```
    print(yes / known)
else:
    print("No known responses")
```

The printed **proportion** is 0.75, or 75 percent, among four known responses. Yes among all five supplied responses is 0.6, or 60 percent. Both calculations are correct when labelled; they describe different **denominator**s.

Unknown must not silently become No.

1/2 = 50 percent among known; 1/4 = 25 percent among all supplied. Exit: No. The **denominator** is known volunteers, and participation may not represent all students.

▼ Earlier wording, alternatives and picture transcripts

Explain the mechanism

Input → **Process** → **Output**

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

1. Model A: 7 correct out of 10 → 7/10 = **70%**.

2. Model B: 18 correct out of 20 → 18/20 = 90%.

3. Model B has higher measured **accuracy** on these tests.

4. Still ask: were the test cases fair and were some errors more costly?

Recalculate, retrace, or restate the reasoning in your own representation:

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

Concrete teaching case

Nine of twelve held-out classifications are correct.

Compute **accuracy** and state the **denominator**.

Exact explanation to model

9/12=3/4=75%. **Count** all scored cases, including errors.

Explain the added worked example

Key terms

Term	Meaning
Class imbalance	Unequal numbers of examples in different classes
Majority baseline	A rule that always predicts the most common training label
Class-specific correct fraction	Correct predictions for one reference class divided by the number of examples of that class
Balanced accuracy	The average of the class-specific correct fractions with equal weight for each class

Apply the connected idea: Summaries with counts and proportions

Model the reasoning aloud

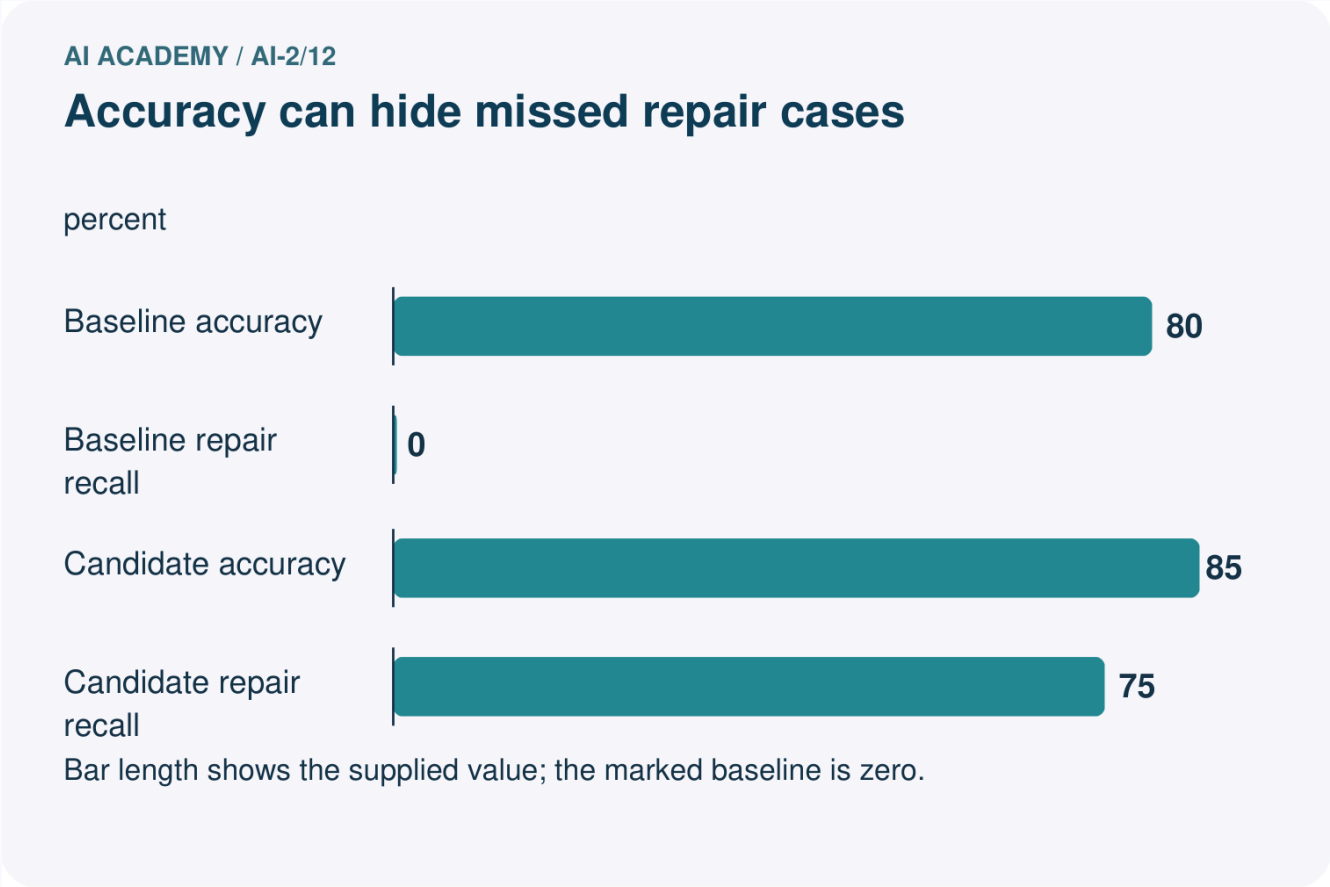
Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

A pictorial worked case: Accuracy as a Fraction

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-2/12

Accuracy as a Fraction

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Reference population

20 labelled cases, including four repair-needed books.



2. Baseline

16/20=80% accuracy; catches 0/4 needed cases.



3. Candidate

(3+14)/20=85% accuracy; catches 3/4=75% needed cases.



4. Compare costs

Candidate also raises two false alarms; discuss both missed repairs and extra checks.

CHANGE ONE THING

A candidate instead marks every book repair needed.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Reference population	20 labelled cases, including four repair-needed books.
2. Baseline	16/20=80% accuracy; catches 0/4 needed cases.
3. Candidate	(3+14)/20=85% accuracy; catches 3/4=75% needed cases.
4. Compare costs	Candidate also raises two false alarms; discuss both missed repairs and extra checks.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Guide denominator choices, then withdraw help

> Collect each learner’s reasoning before revealing keys.

Main W2 question and key

Completion task: For the worked candidate [[12,4],[0,4]], give ordinary **accuracy**, both **class-specific correct fraction**s, **balanced accuracy**, false positives and false negatives. Compare with the baseline.

Teacher check: **Accuracy** 16/20=80 percent. Clear fraction 12/16=75 percent; blocked fraction 4/4=100 percent; **balanced accuracy** (75+100)/2=87.5 percent. FP=4,FN=0.

Baseline **accuracy** is also 80 percent but **balanced accuracy** is 50 percent with FP=0,FN=4.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Challenge the accuracy misconception

Use the worked case for **Accuracy** and **class imbalance**. Proposed explanation: “High **accuracy** means both classes work well”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: A common class can dominate the overall **count**. Inspect each reference-class **denominator** and its errors.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Use the shelf question after the changed scenario

Why is 100% repair detection not sufficient to declare the changed classifier best?

It catches all four repairs but wrongly flags all 16 others, creating many extra checks; consequences must be assessed.

Changed-case reasoning: Recall is 4/4=100%, but **accuracy** is only 4/20=20%. All 16 not-needed books become false alarms.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Keep learners explaining

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

▼ Optional poll teaching cycle

Target: **Count** categories with a stated **denominator**

Reserve further sessions for this topic; the 135-minute primary plan does not absorb all added content.

Completion practice: Poll A has 2 Yes of 4 known and Poll B has 3 Yes of 6 known. Compare **proportion**s and **count**s.

Teacher check: Both are 50 percent. **Count**s and sample sizes differ: 2/4 versus 3/6.

Individual check: For Summaries with **count**s and **proportion**s, use its worked case to evaluate this proposed explanation: "A successful run proves the conclusion". Explain the deciding step.

Expected correction: A **percentage** from a few volunteers is not automatically representative of the whole class. Report **count**s, the question, who responded and exclusions alongside the **percentage**.

Apply the same support, independent transfer and delayed-review decisions to the named connected-topic workbook tasks.

▼ Earlier wording, alternatives and picture transcripts

Demonstrate and guide practice

Board sequence

Write the given facts: Nine of twelve held-out classifications are correct.

Write the question to explain: Compute **accuracy** and state the **denominator**.

Reveal after **prediction**s: 9/12=3/4=75%. **Count** all scored cases, including errors.

Use the earlier two-fraction question as optional practice. Reserve the selected 18-clear/2-blocked comparison for the first independent attempt before feedback.

Classroom dialogue

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and **denominator**s for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

Model these guided questions

Guided: baseline has FP=0,FN=4, finding none of four blocked cases. Candidate has FP=4,FN=0, finding all four but adding four false alarms. Both have sixteen correct **prediction**s.

Equal averaging gives baseline (100+0)/2=50 percent and candidate (75+100)/2=87.5 percent. Exit: request the full labelled matrix, class **count**s, sampling method, **training**-derived baseline and task consequences.

The worked always-clear rule misses all four blocked cases.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start to 7 minutes	Start with a simple correct/all fraction, then introduce matrix orientation and missed-case terminology in this lesson; Week13 has not yet been taught.
7 to 18 minutes	Use counters to score the always-clear baseline and complete W1.
18 to 32 minutes	Compare the candidate matrix. Calculate ordinary and class-specific correct fractions in W2.

Stage	Teacher action
32 to 45 minutes	Trace the code and read the visual. Answer the guided question and explain equal class weighting.
Session 2 start to 8 minutes	Retrieve each denominator and why a high overall score can hide rare-class failure.
8 to 20 minutes	Complete W3 with a changed class mix. Separate the arithmetic assumption from claims about deployment.
20 to 35 minutes	Collect W4 independently before support. Require fractions before rounded percentages and the error tradeoff.
35 to 45 minutes	Use W5 and the exit question to discuss missing classes. Offer the fresh retry after feedback and record support.

Teach the connected idea

Stage	Teacher action
Session 1 start to 7 minutes	Retrieve the prerequisite and explain the purpose in the opening section.
7 to 20 minutes	Read the worked example and diagram; ask learners to trace each step aloud.
20 to 33 minutes	Complete W1 and W2, then compare predictions with observed results.
33 to 45 minutes	Collect the guided prediction before running; discuss a mismatch and its cause.
Session 2 start to 8 minutes	Retrieve the main rule without showing the previous answer.
8 to 20 minutes	Use W3 to diagnose and repair the stated failure; rerun an earlier case.
20 to 35 minutes	Collect W4 independently, including code or plan, predictions and explanations, before hints.
35 to 45 minutes	Discuss W5 and the exit; use the fresh retry when support was needed.

Reduce help deliberately

Individual reasoning check

Connected teaching cycle: Summaries with counts and proportions

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the supplied prediction-record audit

> Keep arithmetic checks separate from future-performance claims.

Supplied lab and answer guidance

The supplied record tallies baseline [[16,0],[4,0]] and candidate [[12,4],[0,4]]. Both ordinary accuracies are16/20=80%. Baseline class fractions100% and0% average50%; candidate75% and100% average87.5%. Baseline FP0/FN4; candidate FP4/FN0. On the separate8/8 assumption, ordinary scores50%/87.5% and balanced scores50%/87.5%. The code raises ValueError if a reference row has zero count; ordinary accuracy can still be calculated separately on observed examples. Do not expose the selected W4 key while modelling; poll-extension feedback has its own task labels.

Predict, tally and inspect errors

1. Predict → Test → Record → Explain

2. Keep expected and actual results separate. Surprises are useful data when recorded honestly.

3. Use the supplied twenty-case **prediction** record to tally both matrices, calculate ordinary and balanced accuracy, and inspect false alarms and missed cases. Keep the independently assessed 18-clear/2-blocked case unused during shared practice.

4. What did the hardest test teach you that the normal test did not?

Keep the record

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose the main misconceptions

High **accuracy** means both classes work well: A common class can dominate the overall **count**. Inspect each reference-class **denominator** and its errors.

Balanced accuracy fixes the model: It changes the summary weighting, not any **prediction** or underlying error.

An absent class has zero measured performance: Its fraction has a zero **denominator** and is unavailable. Report missing coverage rather than inventing a value.

Choose support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ Optional poll misconceptions

A successful run proves the conclusion: A **percentage** from a few volunteers is not automatically representative of the whole class. Report **count**s, the question, who responded and exclusions alongside the **percentage**.

Matching the example is enough: Require the new W4 case, explicit **prediction**s and an explanation. Fresh retry: No, Yes, Unknown, No, Unknown gives 1 Yes of 3 known, about 33.3 percent, with 2 unknown.

Accept 1/3 without rounding.

A convenient shortcut preserves meaning: Dividing by known when it equals 0 causes ZeroDivisionError. Test known > 0 before division and report no known responses if none exist.

Reporting 0 percent would wrongly suggest known responses with no Yes answers.

▼ Earlier wording, alternatives and picture transcripts

Run the weekly evidence lab

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Make the lab runnable

Use the supplied20-case prediction audit for shared practice. Reserve the selected18-clear/2-blocked case for the independent attempt; the poll extension uses separate named tasks and feedback.

Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

Check these misconceptions

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the selected 18-and-2 comparison

> Keep this key until after the first independent attempt.

Selected W4 and checked answer

Selected case: New **training** labels have a clear majority. A separate evaluation has 18 clear and 2 blocked records.

The always-clear baseline is [[18,0],[2,0]] and candidate is [[15,3],[1,1]], with reference rows and clear then blocked on both axes. Calculate ordinary **accuracy**, both class fractions and **balanced accuracy** for each.

Give candidate FP and FN, then explain why the metrics do not choose an automatic winner.

Baseline **accuracy** 18/20=90 percent; class fractions 100 percent and 0 percent; **balanced accuracy** 50 percent.

Candidate **accuracy** 16/20=80 percent; clear fraction 15/18=5/6 about 83.3 percent, blocked fraction 1/2=50 percent; **balanced accuracy** (5/6+1/2)/2=2/3 about 66.7 percent. Candidate FP=3,FN=1.

It finds one blocked case but adds three false alarms; task consequences and the small sample matter.

Judge explanation and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give a repair action

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 baseline key

Training has 16 clear and 4 blocked labels. State the majority rule. On the worked evaluation, give its matrix, **accuracy**, false positives and false negatives with blocked positive. Explain why 80 percent is incomplete.

The **training**-majority rule always predicts clear. Matrix [[16,0],[4,0]] gives 16/20=80 percent, FP=0,FN=4. It misses all four blocked cases, so the overall score hides complete failure on that evaluated class.

W2 candidate key

For the worked candidate [[12,4],[0,4]], give ordinary **accuracy**, both **class-specific correct fraction**s, **balanced accuracy**, false positives and false negatives. Compare with the baseline.

Accuracy 16/20=80 percent. Clear fraction 12/16=75 percent; blocked fraction 4/4=100 percent; **balanced accuracy** (75+100)/2=87.5 percent. FP=4,FN=0.

Baseline **accuracy** is also 80 percent but **balanced accuracy** is 50 percent with FP=0,FN=4.

W3 changed-mix key

For the illustrative 8-clear 8-blocked set, use baseline [[8,0],[8,0]] and candidate [[6,2],[0,8]]. Calculate ordinary and **balanced accuracy** for each. Explain what stayed fixed and what this example does not prove.

Baseline ordinary 8/16=50 percent and balanced (100+0)/2=50 percent. Candidate ordinary 14/16=87.5 percent and balanced (75+100)/2=87.5 percent. Within-class fractions stayed fixed while class **count**s changed.

This arithmetic assumption does not prove unchanged real-world behavior on a new population.

W5 absent-class key

An evaluation contains ten clear cases and no blocked cases. All **prediction**s are clear. State ordinary **accuracy**, explain why blocked performance is unavailable, and describe how our code handles the missing class.

Ordinary **accuracy** is 10/10=100 percent on the observed clear cases. The blocked-class fraction has **denominator** zero, so it cannot be measured.

Our two-class summary function stops with ValueError; report the missing coverage instead of claiming perfect performance on both classes.

▼ Poll-extension questions and keys

Summaries with counts and proportions — W1 State the denominator

In the worked poll, why is the known **denominator** 4 rather than 5?

Only Yes and No are included; one Unknown is excluded and must be disclosed.

Summaries with counts and proportions — W2 Compare proportions

Poll A has 2 Yes of 4 known and Poll B has 3 Yes of 6 known. Compare **proportion**s and **count**s.

Both are 50 percent. **Count**s and sample sizes differ: 2/4 versus 3/6.

Summaries with counts and proportions — W3 Repair zero division

All three responses are Unknown. What should the program report?

No known responses, with 3 unknown. Do not divide by zero or report 0 percent as a measured **proportion**.

Summaries with counts and proportions — W4 Summarize independently

Process Yes, Yes, No, Unknown, No, Yes. Report Yes **count**, known **count**, unknown **count** and Yes **proportion** among known.

3 Yes, 5 known, 1 unknown; 3/5 = 0.6 = 60 percent.

Summaries with counts and proportions — W5 Avoid overclaiming

Three of five known volunteers said Yes. Can you say 60 percent of all students agree?

No. The **denominator** is known volunteers, and participation may not represent all students.

▼ Earlier wording, alternatives and picture transcripts

Assess and explain the answers

Worked case answer

9/12=3/4=75%. **Count** all scored cases, including errors.

Additional fraction-comparison practice

A classifier gets 9 of 12 cases right. Another gets 8 of 10 right. Calculate both accuracies and identify the larger fraction.

Reasoned practice answer

9/12 is 75%; 8/10 is **80%**. The second fraction is higher, although differing test sets can make a direct model comparison unfair.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not **count** a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “More correct answers always means higher **accuracy**.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

A=**80%**, B=**60%**; combined 26/40=65%. The unweighted average is **70%**, which gives the ten-case class the same weight as the thirty-case class. State whether the question concerns a typical class or a typical tested case.

Explanations for the additional workbook examples

W1 Score the majority baseline

W2 Compare both class fractions

W3 Explain a changed class mix

W4 Assess a new comparison independently

New **training** labels have a clear majority. A separate evaluation has 18 clear and 2 blocked records.

The always-clear baseline is [[18,0],[2,0]] and candidate is [[15,3],[1,1]], with reference rows and clear then blocked on both axes. Calculate ordinary **accuracy**, both class fractions and **balanced accuracy** for each.

Give candidate FP and FN, then explain why the metrics do not choose an automatic winner.

Baseline **accuracy** 18/20=90 percent; class fractions 100 percent and 0 percent; **balanced accuracy** 50 percent.

Candidate **accuracy** 16/20=80 percent; clear fraction 15/18=5/6 about 83.3 percent, blocked fraction 1/2=50 percent; **balanced accuracy** (5/6+1/2)/2=2/3 about 66.7 percent. Candidate FP=3,FN=1.

It finds one blocked case but adds three false alarms; task consequences and the small sample matter.

W5 Limit claims from missing classes

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect the score, support and delayed return

> Preserve the declared evaluation unit and all denominators.

Portfolio evidence

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Compare recommender choices with a simple baseline and report relevant rare-category failures, not only aggregate **accuracy**.

Support and fresh main assessment

Use **count**ers and write each fraction with its reference-class total before converting to a **percentage**. Supply blank **denominator** boxes rather than answers.

Extension: express ordinary **accuracy** as the class-size-weighted average of the two correct fractions; for the candidate use (16/20)(12/16)+(4/20)(4/4)=16/20.

Require W4 baseline ordinary 90 percent and balanced 50 percent; candidate ordinary 80 percent and balanced 2/3 about 66.7 percent, with FP=3,FN=1.

Fresh retry: evaluation 12 clear and 4 blocked, baseline [[12,0],[4,0]], candidate [[9,3],[1,3]], **training** majority clear. Baseline ordinary 75 percent, class fractions 100 and 0 percent, balanced 50 percent.

Candidate ordinary 12/16=75 percent, both class fractions 75 percent, balanced 75 percent, FP=3,FN=1. Require both **denominator**s and no automatic-winner claim.

Schedule a changed-case return

Suggested return points in this level: ai-2/13, ai-2/15, ai-2/18. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	

Record	Teacher evidence
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Bridge to next week

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Next: Week 13: Types of Classification Error. Retrieve today's **evidence** and identify what carries forward and what is new.

Next is Week13: Types of Classification Error. Carry forward the matrix orientation, positive class and error counts. Teacher reference: https://scikit-learn.org/stable/modules/generated/sklearn.metrics.balanced_**accuracy**_score.html (equal average of class recalls; checked2026-10-17).

▼ Optional poll support and reassessment

Use fraction bars with separate unknown cards. Extension: show both **denominator**s side by side with explicit labels.

Independent W4 **evidence**: 3 Yes, 5 known, 1 unknown; 3/5 = 0.6 = 60 percent. Record whether the learner needed no help, a hint or explanation.

Fresh retry: No, Yes, Unknown, No, Unknown gives 1 Yes of 3 known, about 33.3 percent, with 2 unknown. Accept 1/3 without rounding.

▼ Earlier wording, alternatives and picture transcripts

Connect and extend learning

Transparent Book Recommender

Report correct recommendations / all test users, alongside the raw test **count**.

Support and stretch

Trace one record or condition at a time with **count**ers and labelled columns. Stretch: introduce a tie, boundary or missing field and declare its handling policy before rerunning.

Exit reflection

Bridge to the next week

Support and fresh reassessment

Readiness and reassessment for Summaries with counts and proportions

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative teaching checklist**

REFERENCE · SECTION 7

Reference · Alternative teaching checklist

> Use alongside the main route when helpful.

Planning reference

Teaching target: Explain why overall **accuracy** can hide failure on a less common class

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI ac**count** is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

Before class, solve the supported practice: For the worked candidate [[12,4],[0,4]], give ordinary **accuracy**, both **class-specific correct fraction**s, **balanced accuracy**, false positives and false negatives.

Compare with the baseline.

Reasoned practice answer: **Accuracy** 16/20=80 percent. Clear fraction 12/16=75 percent; blocked fraction 4/4=100 percent; **balanced accuracy** (75+100)/2=87.5 percent. FP=4,FN=0.

Baseline **accuracy** is also 80 percent but **balanced accuracy** is 50 percent with FP=0,FN=4.

Check every learner before discussion: Use the worked case for **Accuracy** and **class imbalance**. Proposed explanation: “High **accuracy** means both classes work well”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: A common class can dominate the overall **count**. Inspect each reference-class **denominator** and its errors.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: A candidate instead marks every book repair needed. Require a **prediction** and a reason before comparing. Expected result: Recall is 4/4=100%, but **accuracy** is only 4/20=20%.

Reason: All 16 not-needed books become false alarms.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback and saved evidence**

REFERENCE · SECTION 8

Reference · Feedback and saved evidence

> Use alongside the main route when helpful.

Planning reference

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.
</details>

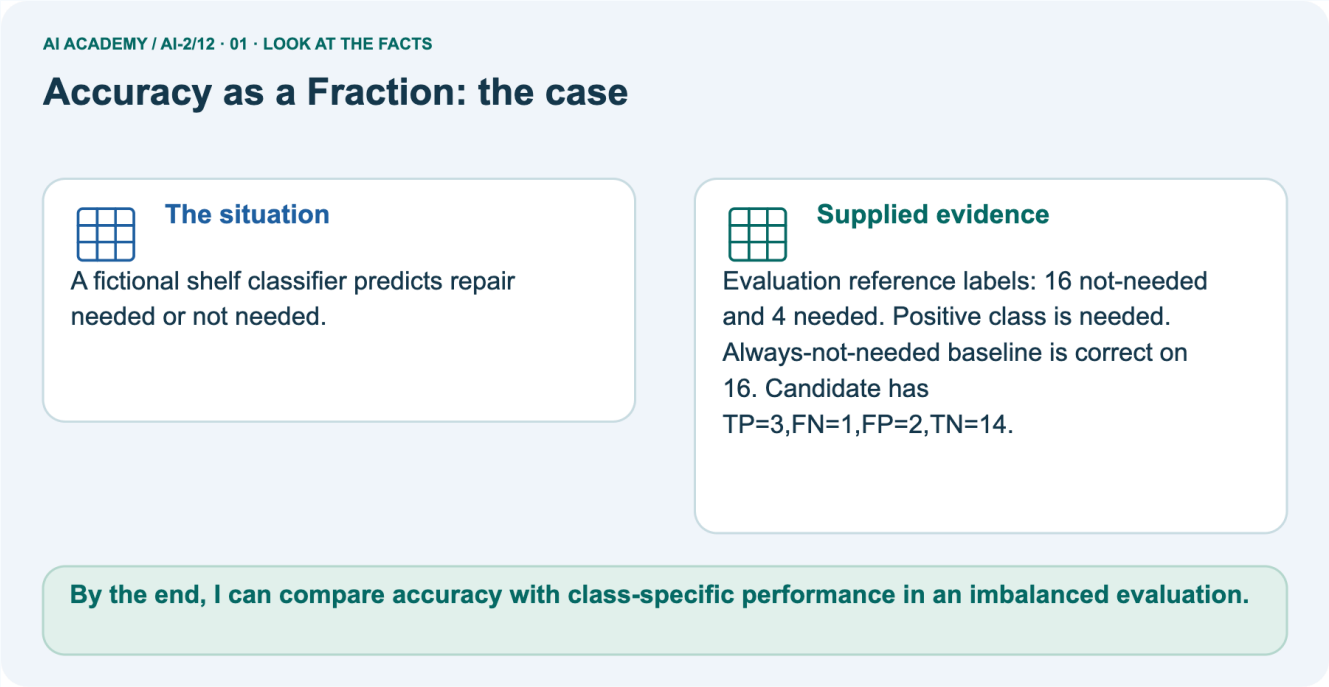
<details>
<summary>Reference · Teach with the six pictures</summary>
</details>

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose a picture for a specific explanation.

01. Accuracy as a Fraction: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

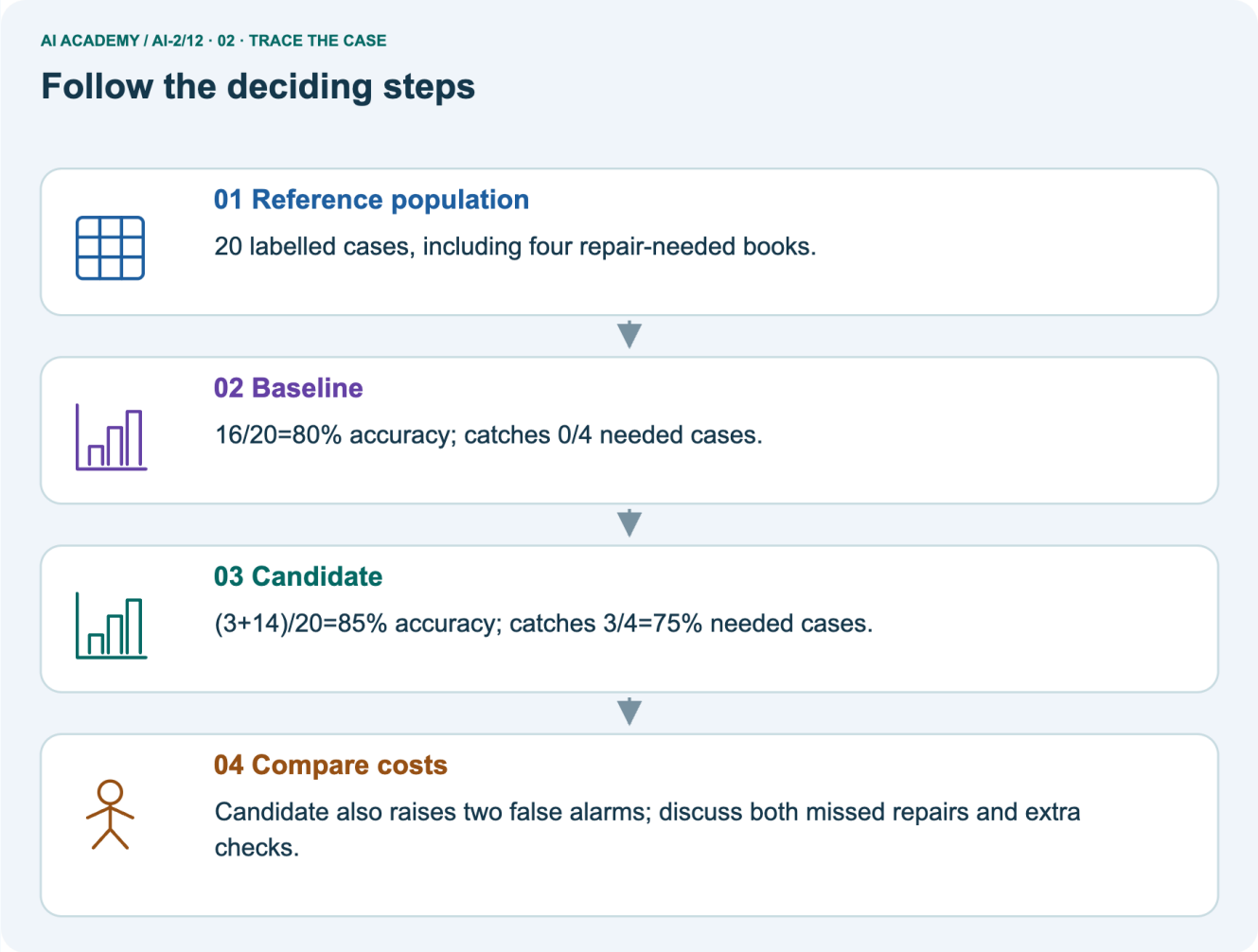
A fictional shelf classifier predicts repair needed or not needed. Supplied facts: Evaluation reference labels: 16 not-needed and 4 needed. Positive class is needed. Always-not-needed baseline is correct on 16.

Candidate has TP=3,FN=1,FP=2,TN=14.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



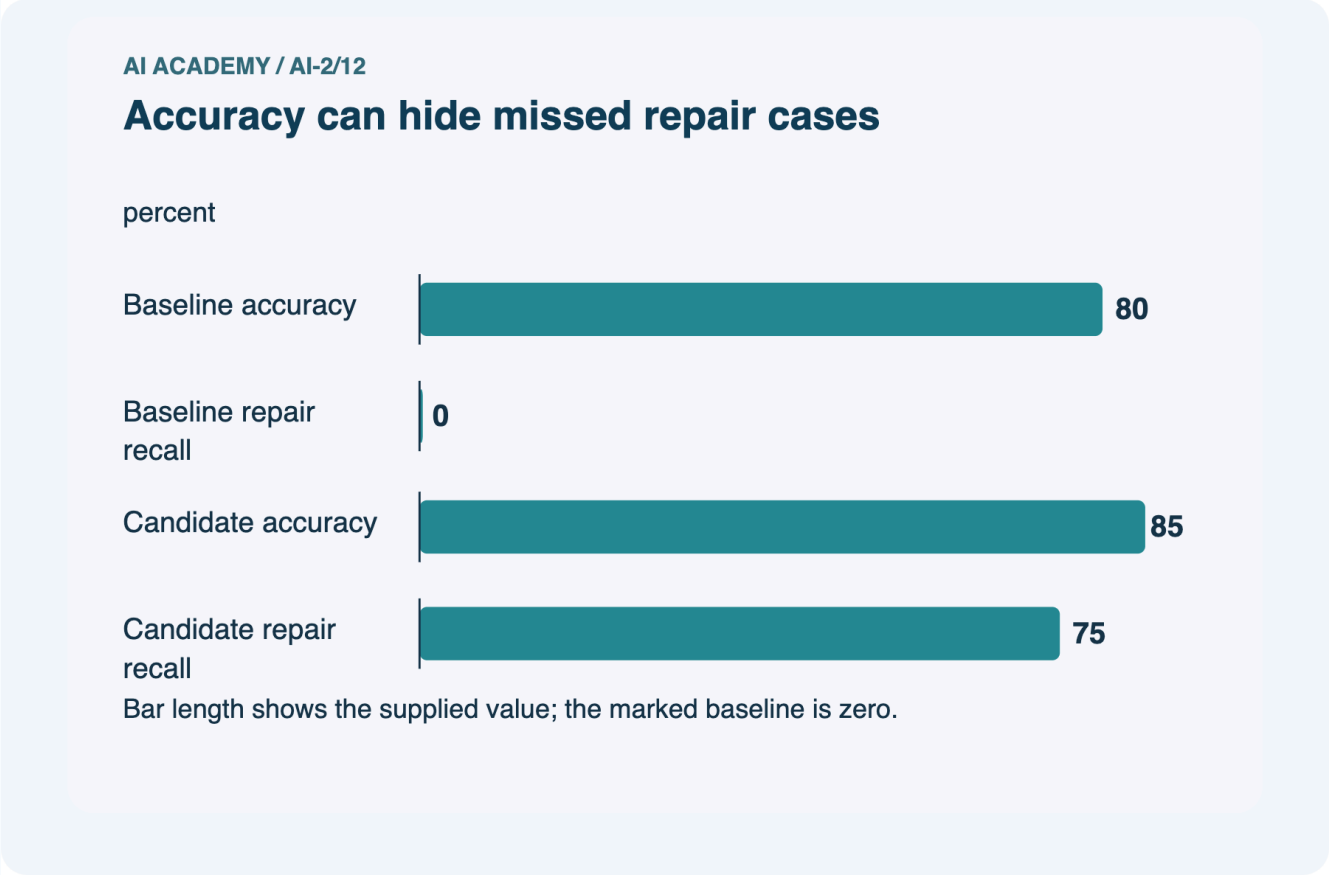
Follow the labelled picture with the matching explanation. Open the vector diagram

Reference population: 20 labelled cases, including four repair-needed books. → Baseline: 16/20=**80% accuracy**; catches 0/4 needed cases. → Candidate: (3+14)/20=85% **accuracy**; catches 3/4=75% needed cases. → Compare costs: Candidate also raises two false alarms; discuss both missed repairs and extra checks.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Accuracy can hide missed repair cases



Follow the labelled picture with the matching explanation. Open the vector diagram

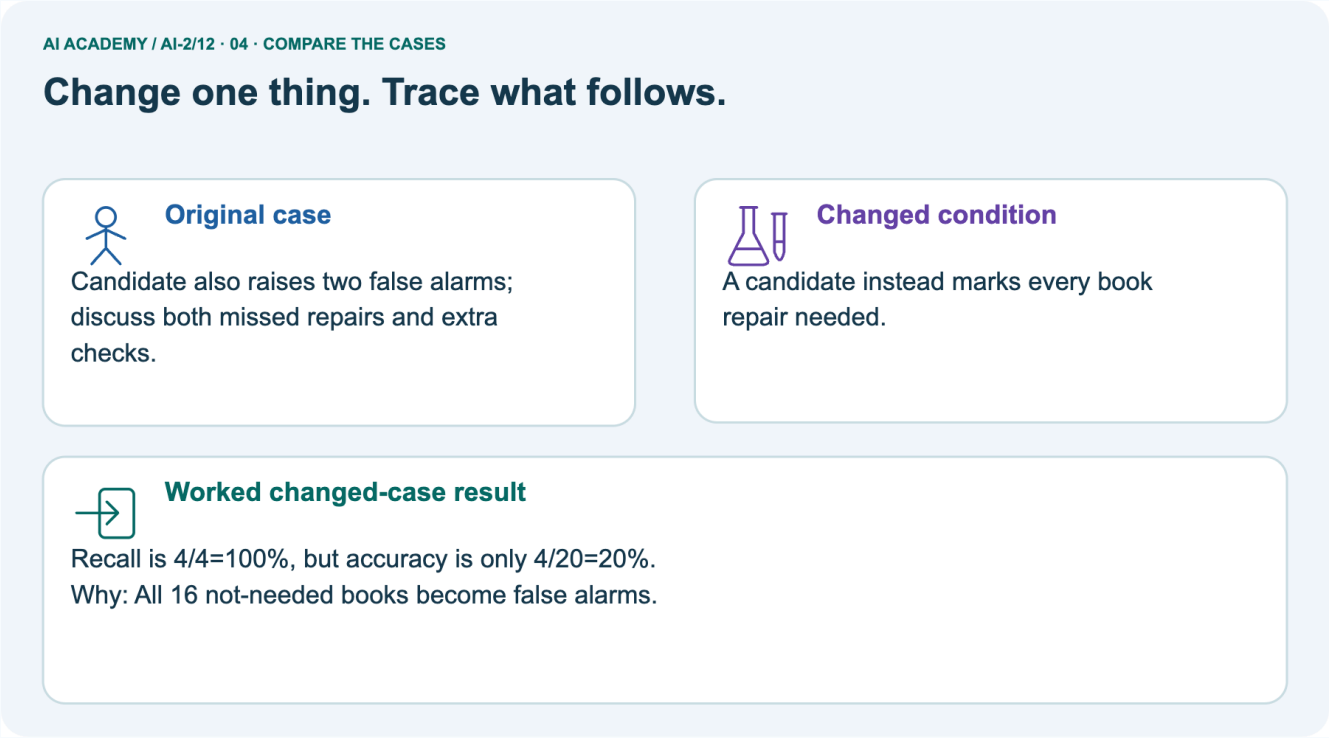
Accuracy can hide missed repair cases. Units: percent. Baseline **accuracy**: 80. Baseline repair recall: 0. Candidate **accuracy**: 85. Candidate repair recall: 75.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and **denominator**s before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Candidate also raises two false alarms; discuss both missed repairs and extra checks. Changed condition: A candidate instead marks every book repair needed.

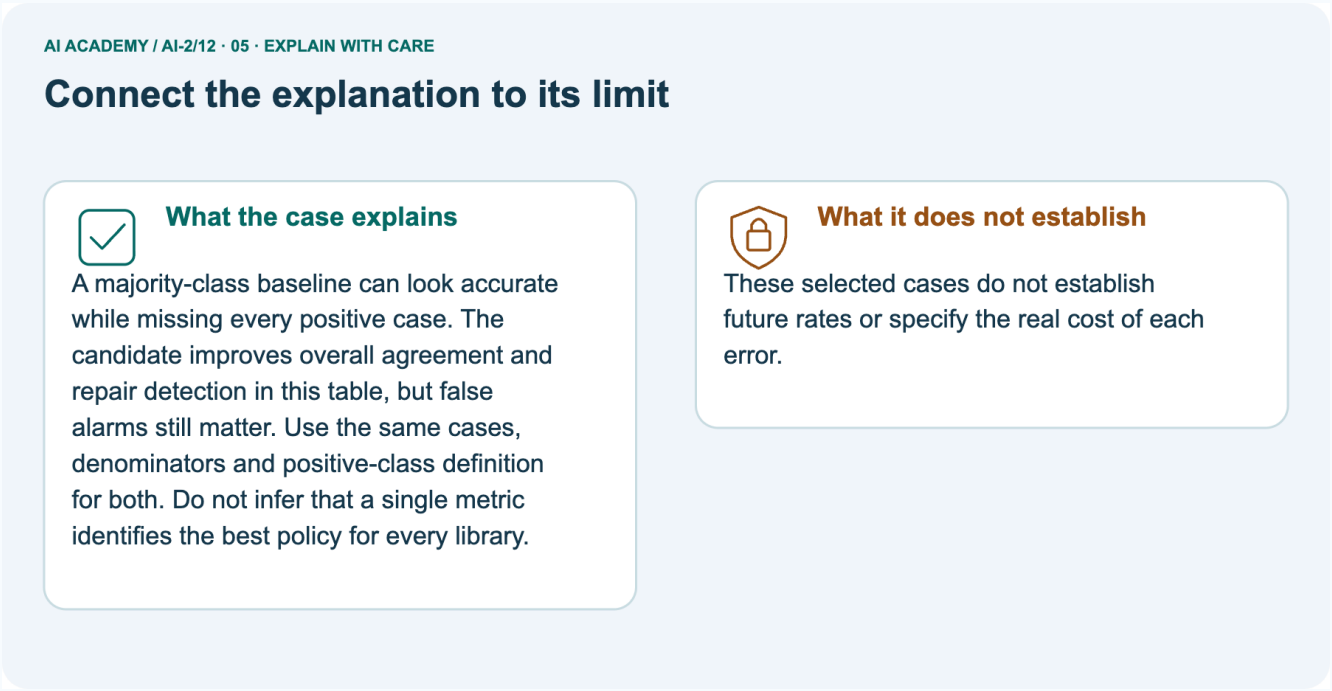
Expected result: Recall is 4/4=100%, but **accuracy** is only 4/20=20%. Reason: All 16 not-needed books become false alarms.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: All 16 not-needed books become false alarms.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A majority-class baseline can look accurate while missing every positive case. The candidate improves overall agreement and repair detection in this table, but false alarms still matter.

Use the same cases, **denominator**s and positive-class definition for both. Do not infer that a single metric identifies the best policy for every library.

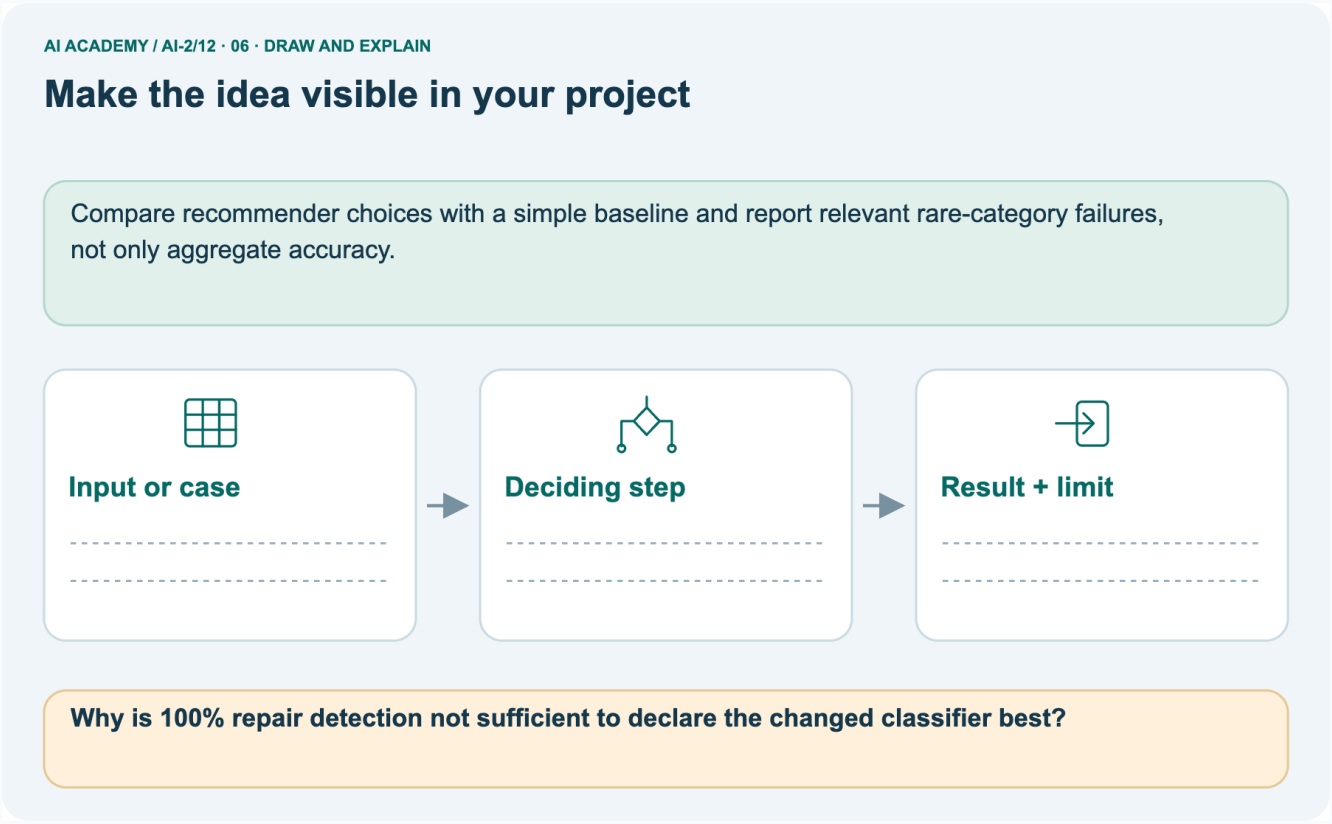
Limit: These selected cases do not establish future rates or specify the real cost of each error.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: These selected cases do not establish future rates or specify the real cost of each error.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Compare recommender choices with a simple baseline and report relevant rare-category failures, not only aggregate **accuracy**. Draw your **input**, deciding step, result and limit.

Supported question: Why is 100% repair detection not sufficient to declare the changed classifier best?

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: It catches all four repairs but wrongly flags all 16 others, creating many extra checks; consequences must be assessed.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Accuracy as a Fraction: the case

02. Follow the deciding steps

03. Accuracy can hide missed repair cases

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.